VICTORY 2027

Climate

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

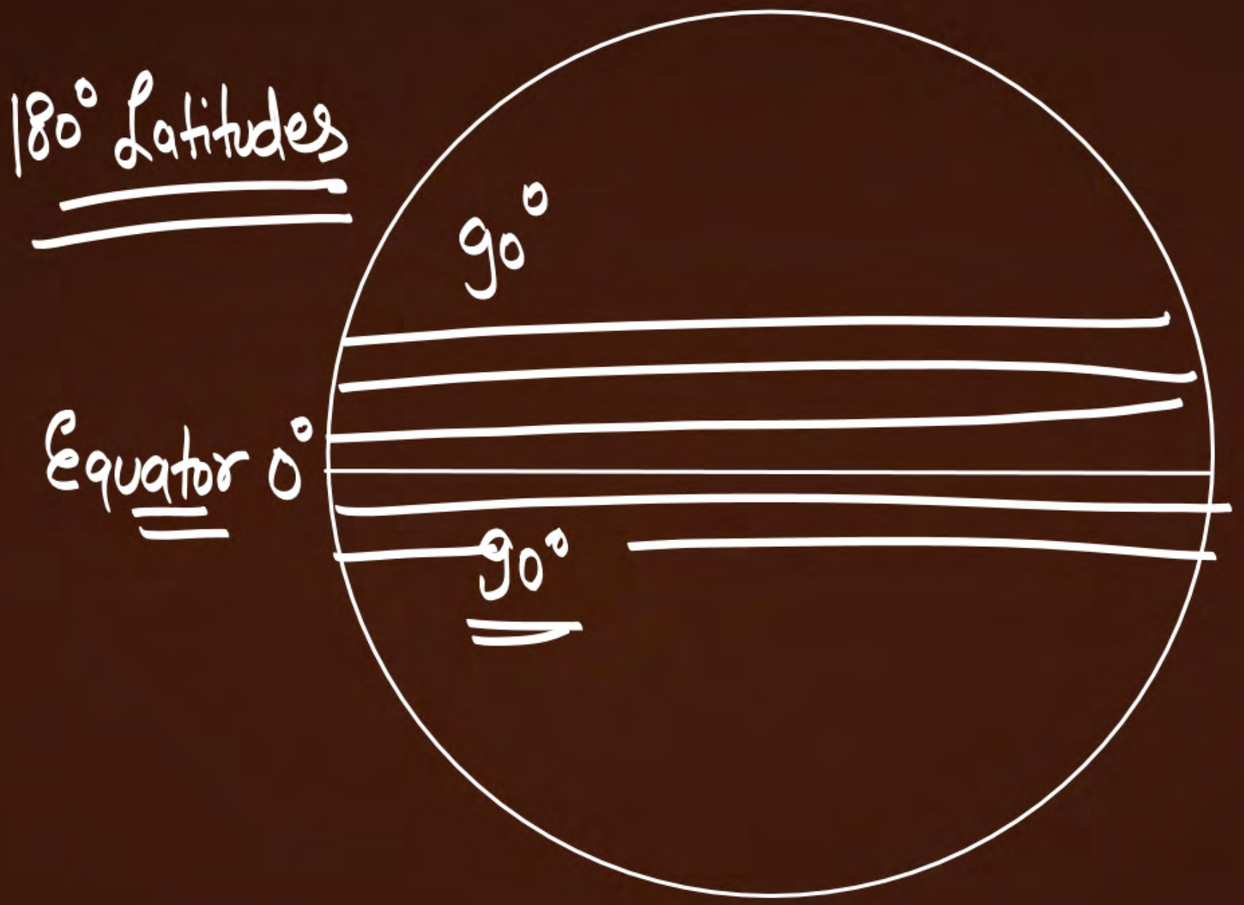
1. Distribution of Temperature
2. Factors affecting the climate of the area
3.
4.

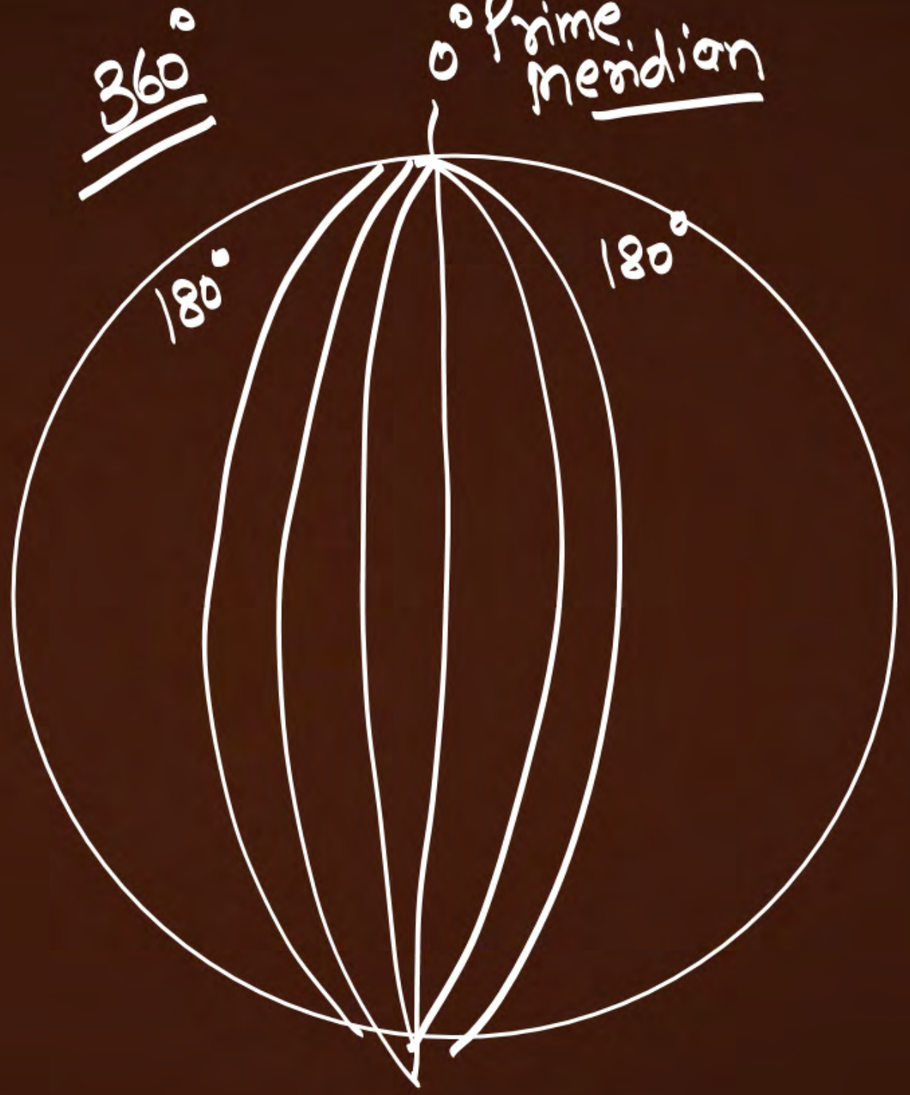
Climate

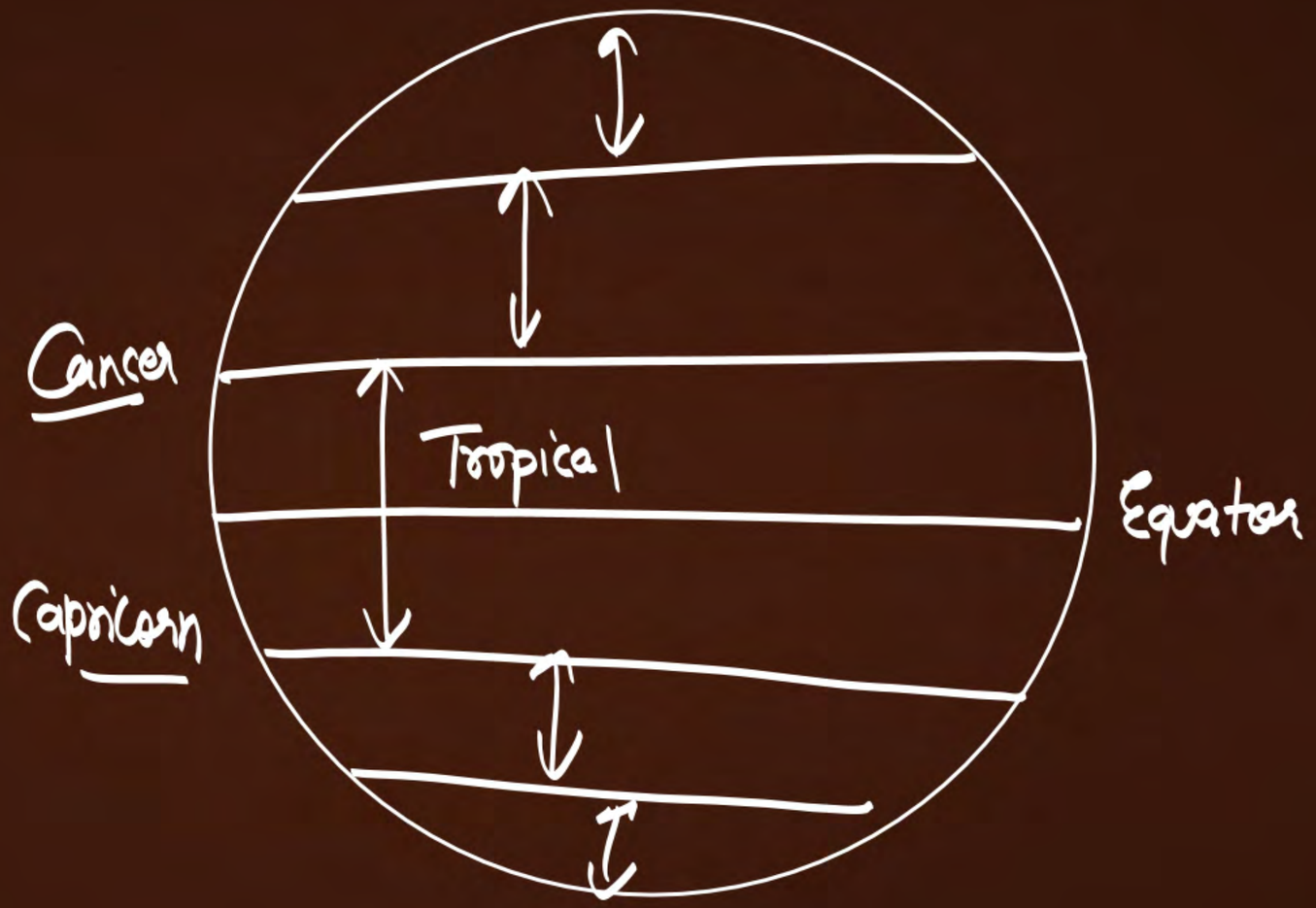
Distribution of Temperature, Rainfall, winds in Summer and Winter and factors affecting the climate of India (latitudes, altitude, distance from sea, winds, Himalayas, jet streams, and El Nino). Monsoon and its mechanism. Seasons: Summer; Monsoon, Retreating Monsoon and Winter. Temperature and pressure conditions, wind movements, local winds, precipitation and interpretation of climate data.

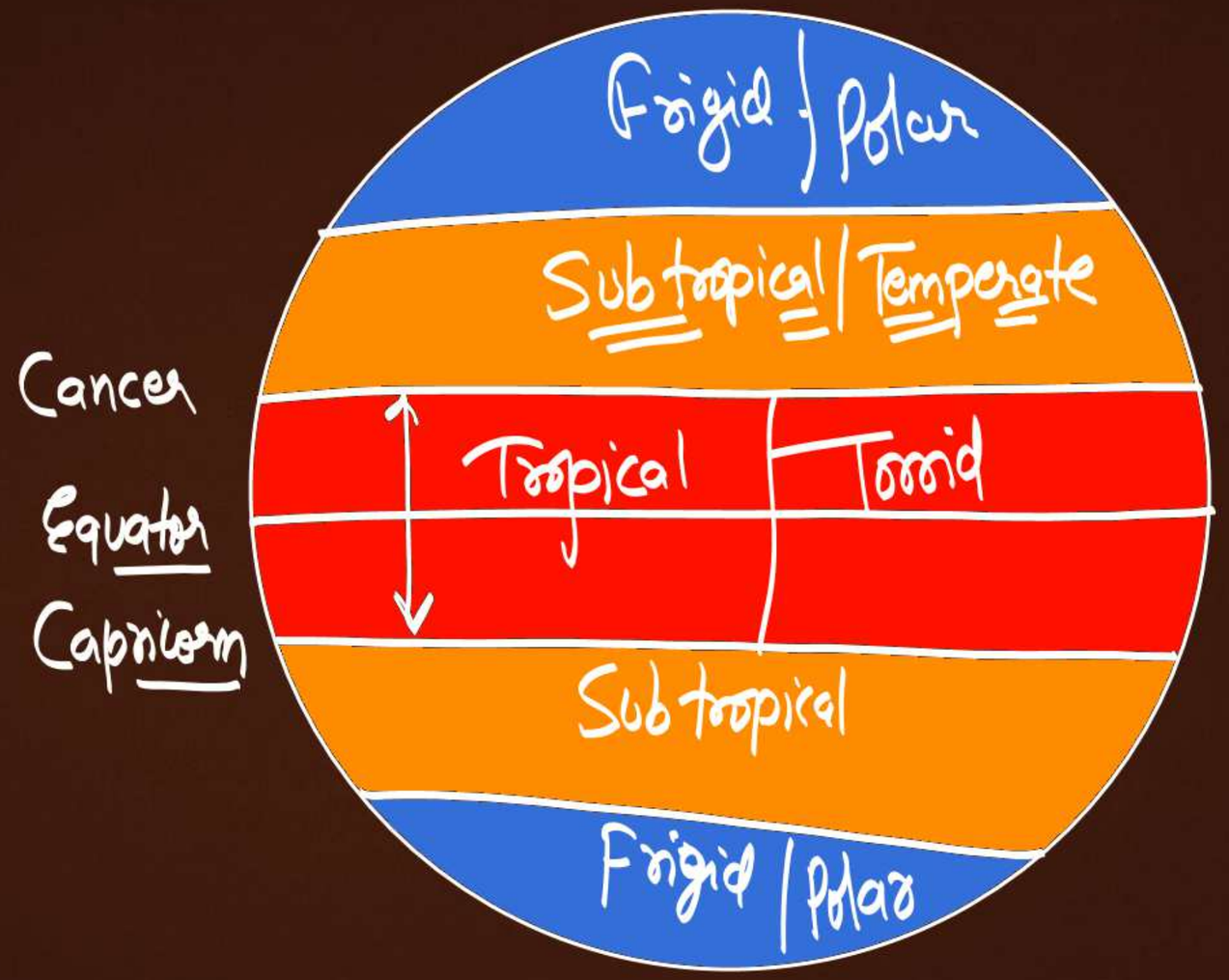
Weather → Temporary atmospheric Condition a place.

Climate → Constant weather Conditions of a place over a long period of time.

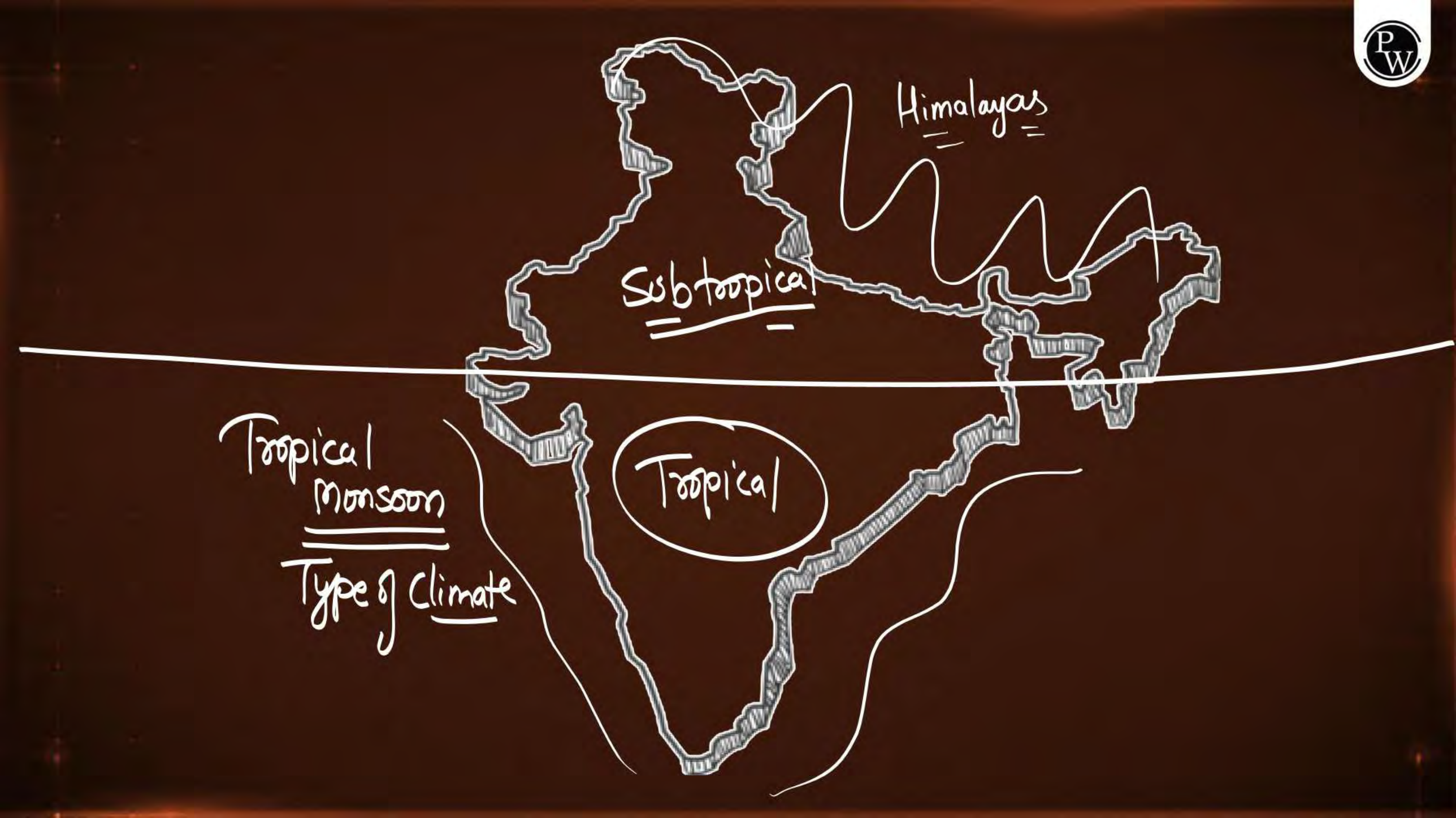

180° Latitudes
90°
Equator 0°
90°


360°
0° Prime meridian
180°
180°


Cancer
Tropical
Equator
Capricorn


Frigid / Polar
Subtropical / Temperate
Cancer
Tropical
Torrid
Equator
Capricorn
Subtropical
Frigid / Polar

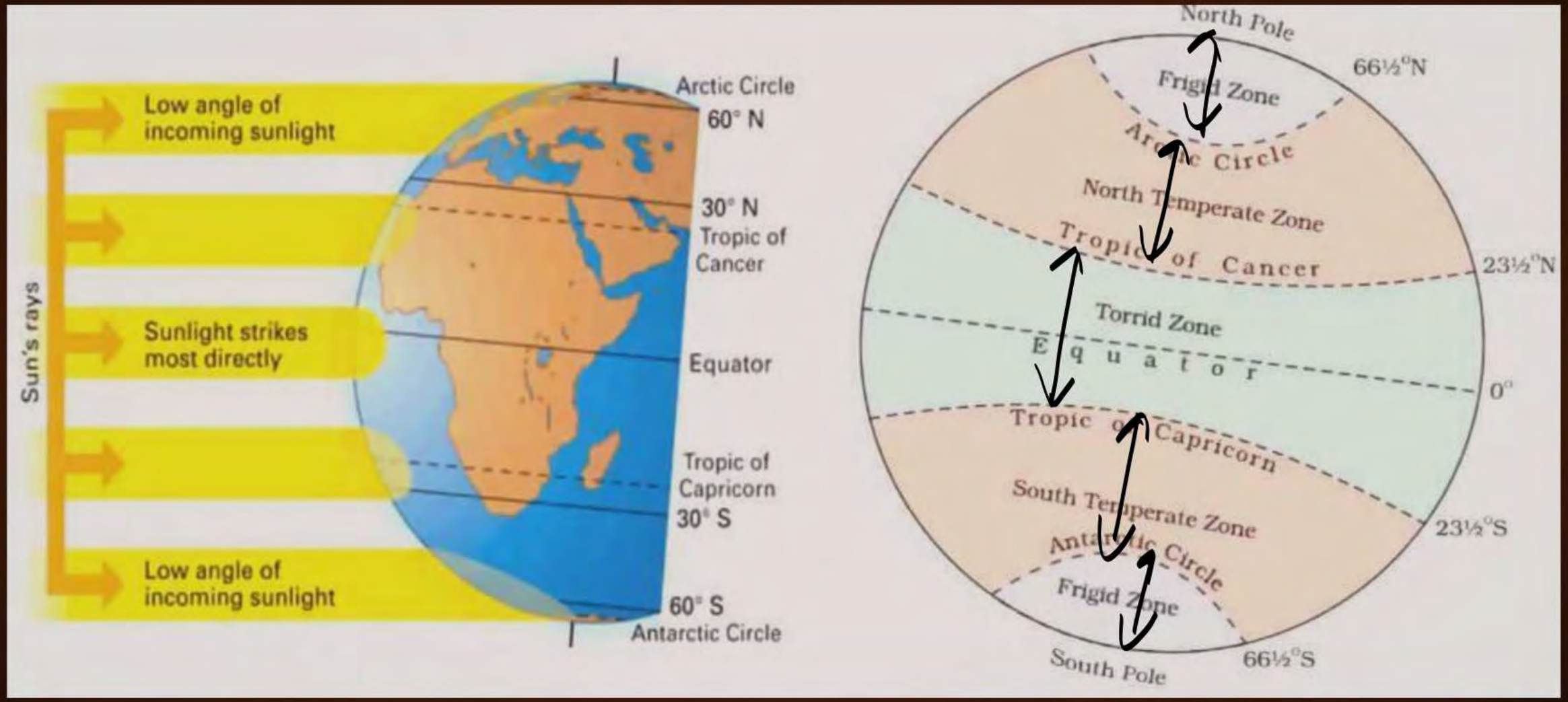

ARCTIC OCEAN
Greenland
(Denmark)
Svalbard
(Norway)
ARCTIC OCEAN
Cold
Russia
United States
Canada
United States
Iceland
Norway
Sweden
Finland
U.K.
Ireland
Estonia
Latvia
Lithuania
Kazakhstan
Mongolia
ATLANTIC OCEAN
Portugal
Spain
France
Italy
Greece
Turkey
Cyprus
Iran
China
N. Korea
S. Korea
Japan
Mexico
Caribbean
Morocco
Tunisia
Algeria
Libya
Egypt
Saudi Arabia
India
PACIFIC OCEAN
Western Sahara
Senegal
Mali
Niger
Chad
Sudan
Eritrea
Djibouti
Yemen
Laos
Vietnam
Taiwan
Philippines
Cambodia
Thailand
Malaysia
Sri Lanka
PACIFIC OCEAN
Central America
Guyana
Suriname
French Guiana (France)
Venezuela
Colombia
Ecuador
Peru
Brazil
Bolivia
Paraguay
Chile
Argentina
Uruguay
Nigeria
Ghana
Togo
Benin
Cote d'Ivoire
Liberia
Sierra Leone
Guinea
Guinea-Bissau
The Gambia
Burkina Faso
Cameroon
C.A.R.
South Sudan
Ethiopia
Somalia
Kenya
Uganda
Rwanda
Burundi
Tanzania
Congo
Gabon
Equatorial Guinea
Angola
Zambia
Zimbabwe
Namibia
Botswana
South Africa
Lesotho
eSwatini
Malawi
Mozambique
Madagascar
ATLANTIC OCEAN
INDIAN OCEAN
Papua New Guinea
Solomon Islands
Timor-Leste
Indonesia
Australia
New Caledonia
Fiji
New Zealand
SOUTHERN OCEAN
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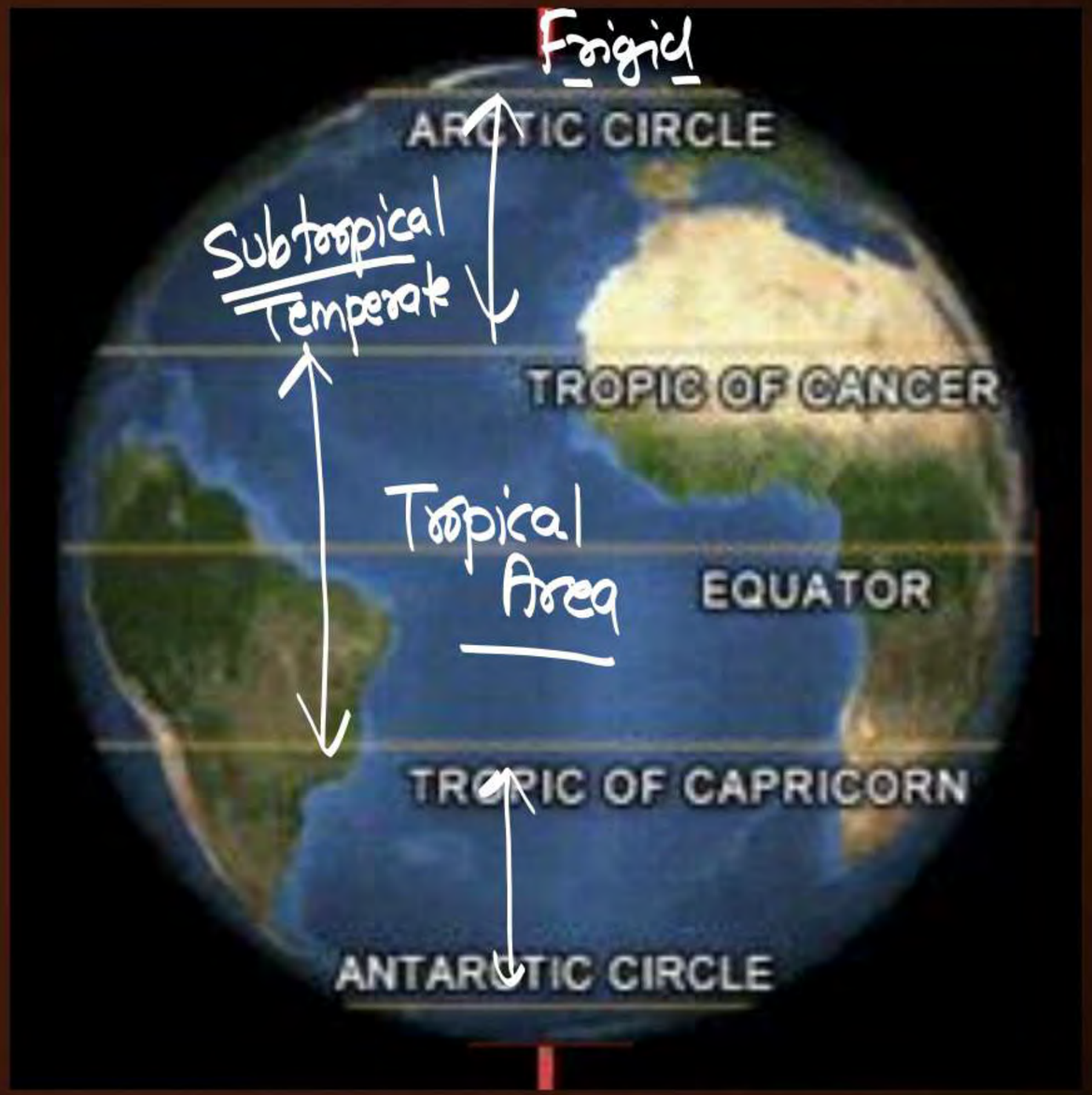

Himalayas
Subtropical
Tropical
Tropical Monsoon Type of Climate

Climate

Monsoon
↓
Mausim
Seasonal movement of winds

- India has a tropical monsoon type of climate.
- This is because India lies in the tropical belt and its climate is influenced by the monsoon winds which are largely confined to the Tropics, i.e., between 20◦ N and 20◦ S.
- The main characteristics of this type of climate are relatively high temperatures and dry winters.
- However, the Himalayas in the north and the Indian Ocean in the south provide distinctive climatic conditions to India.


Low angle of incoming sunlight
Sunlight strikes most directly
Low angle of incoming sunlight
Sun's rays
Arctic Circle
60° N
30° N
Tropic of Cancer
Equator
Tropic of Capricorn
30° S
60° S
Antarctic Circle
North Pole
66½°N
Frigid Zone
Arctic Circle
North Temperate Zone
Tropic of Cancer
23½°N
Torrid Zone
Equator
0°
Tropic of Capricorn
23½°S
South Temperate Zone
Antarctic Circle
Frigid Zone
66½°S
South Pole


Frigid
ARCTIC CIRCLE
Subtropical
Temperate
TROPIC OF CANCER
Tropical
Area
EQUATOR
TROPIC OF CAPRICORN
ANTARCTIC CIRCLE

Climate

- The Himalayan ranges protect northern India from the cold winds of Central Asia and Siberia and give it a continental climate, the characteristics of which are the prevalence of land winds, dryness of air and large diurnal range of temperature.
- The Indian Ocean in the south gives it a hot monsoon climate more typical of the tropical than of the temperate zone.

Regional Variations

- Despite the broad unity of the monsoon type of climate, variations in climate occur in different regions of the country.
- For example, the climatic conditions of Bihar and Uttar Pradesh in the north differ from that of Kerala and Tamil Nadu in the south; yet all of these States have a similar monsoon type of climate.
- These variations are expressed in the pattern of winds, temperature and rainfall, rhythm of seasons and the degree of wetness or dryness.
- These regional diversities are known as the sub-types of monsoon climate.


One Country.
A World of Weather.

Distribution of Temperature

- Seasonal variations in temperature occur from place to place and from region to region.
- Variations in temperature are found also at a single place and in a single day.

Diurnal Range

Difference between Highest & lowest temp. of a place in a singel day.

Bhopal Hi - 40°c. → 15°C
Lo → 25°C.

Barmer - Hi - 45°C → 19°C
- Lo - 26°C.

Chennai Hi - 32°C → 5°C
Lo - 27°C

Manipur → Hi - 29°C → 11°C
→ Low - 18°C

A Climate of Extreme Contradictions



Despite a single climatic label, regional diversities create distinct sub-types of the monsoon climate.

NotebookLM

Distribution of Temperature

Examples of these variations are:

1. Barmer in Rajasthan may record a temperature of 48° C or 50° C on a June day, while it hardly reaches 22° C at Pahalgam or Gulmarg in Kashmir on the same day. However, in Dras near Kargil, the temperatures may go down to -40° C during winter.

2. Kerala has tropical climate with warm and moist air, whereas Punjab has continental climate with severe heat alternating with severe cold.


INDIA
States and Union Territories
INDIA FLAG
TAJIKISTAN
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR (BURMA)
SRI LANKA
LADAKH
Srinagar
Leh
JAMMU & KASHMIR
Jammu
HIMACHAL PRADESH
Shimla
PUNJAB
CHANDIGARH
Dehradun
UTTARAKHAND
HARYANA
DELHI
New Delhi
UTTAR PRADESH
Lucknow
Jaipur
RAJASTHAN
SIKKIM
Gangtok
Itanagar
ARUNACHAL PRADESH
Dispur
ASSAM
NAGALAND
Kohima
Imphal
MANIPUR
MEGHALAYA
BIHAR
Patna
Agartala
TRIPURA
Aizawl
MIZORAM
JHARKHAND
Ranchi
WEST BENGAL
Kolkata
Gandhinagar
GUJARAT
Bhopal
MADHYA PRADESH
CHHATTISGARH
Raipur
ODISHA
Bhubaneswar
BAY OF BENGAL
Daman
DADRA & NAGAR HAVELI AND DAMAN & DIU
MAHARASHTRA
Mumbai
TELANGANA
Hyderabad
Amaravati
Yanam (Puducherry)
ARABIAN SEA
Panaji
GOA
ANDHRA PRADESH
KARNATAKA
Bengaluru
Chennai
Mahe (Puducherry)
PUDUCHERRY
LAKSHADWEEP (INDIA)
Kavaratti
KERALA
TAMIL NADU
Karaikal (Puducherry)
Thiruvananthapuram
Palk Strait
ANDAMAN & NICOBAR ISLANDS (INDIA)
Port Blair
International Boundary
State/UT Boundary
Country Capital
State/UT Capital
Map not to Scale
INDIAN OCEAN
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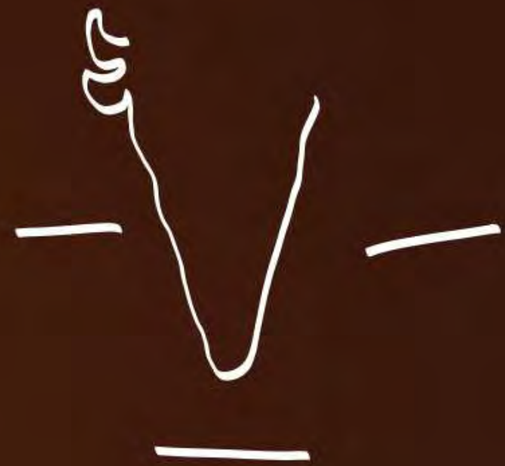
Continentality → A large area surrounded only by land.

Land → gets Hot & Cooled quickly

↓

Night → 20°C → 30°C

Noon → 50°C



Day time

Oceanic influence

Maritime



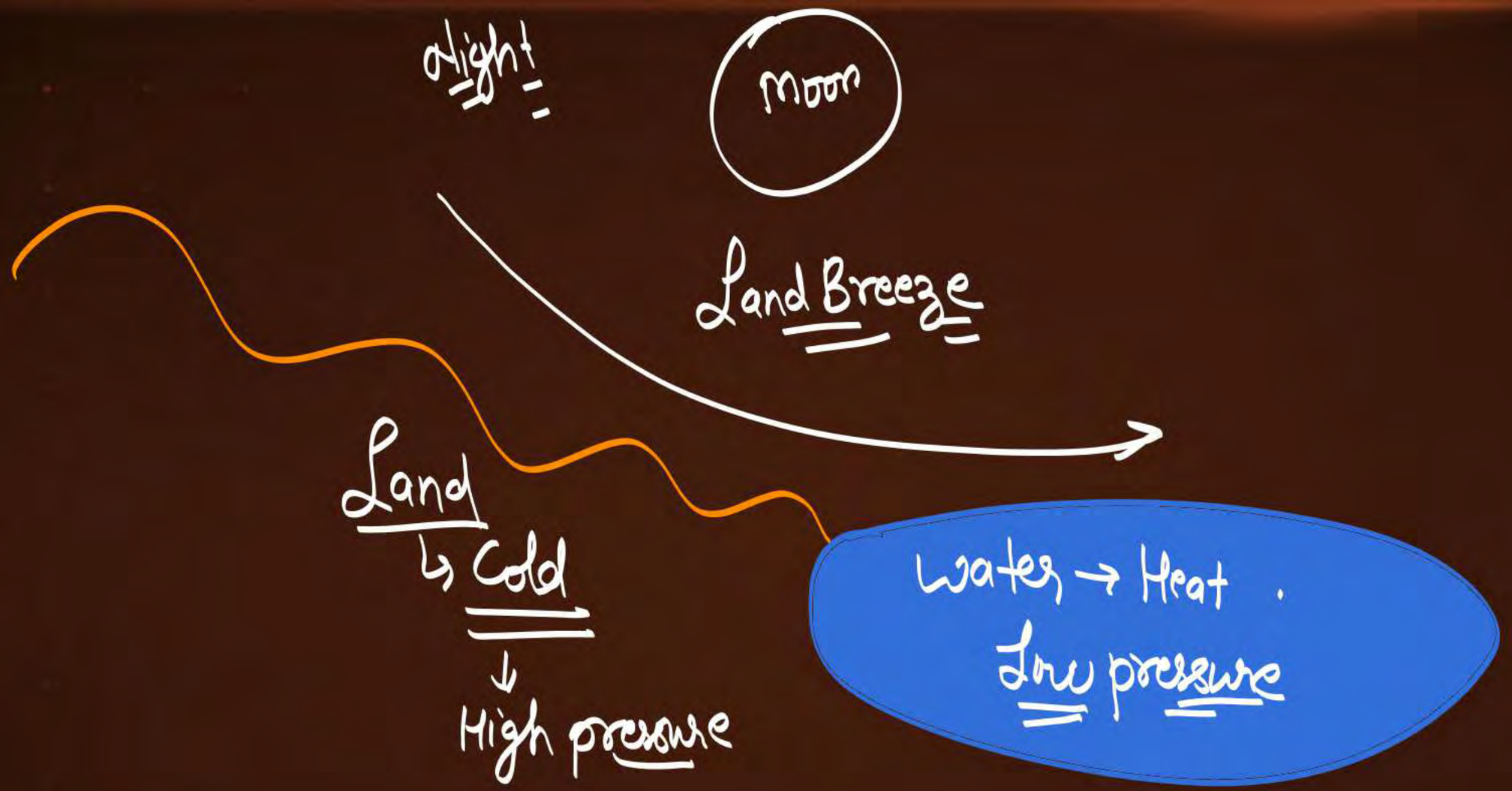
Sea Breeze

Land → Heatup

Low pressure

Water → Cold

High


night
Moon
Land Breeze
Land
Cold
High pressure
Water → Heat
Low pressure


SEA BREEZE (DAYTIME)
SUN
Air moves over sea
WARM, LESS DENSE AIR RISES
LOW PRESSURE
COOLER, DENSER AIR SINKS
HIGH PRESSURE
SEA BREEZE
Cool Air to Land
LAND (HOT)
SEA (COOLER)


LAND BREEZE (NIGHTTIME)
MOON
Air moves over land
COOLER, DENSER AIR SINKS
HIGH PRESSURE
WARMER, LESS DENSE AIR RISES
LOW PRESSURE
LAND BREEZE
Cool Air to Sea
LAND (COOLER)
SEA (WARMER)

Pressure → Air



Heat

↓

Air becomes light



→ Low pressure

Cold

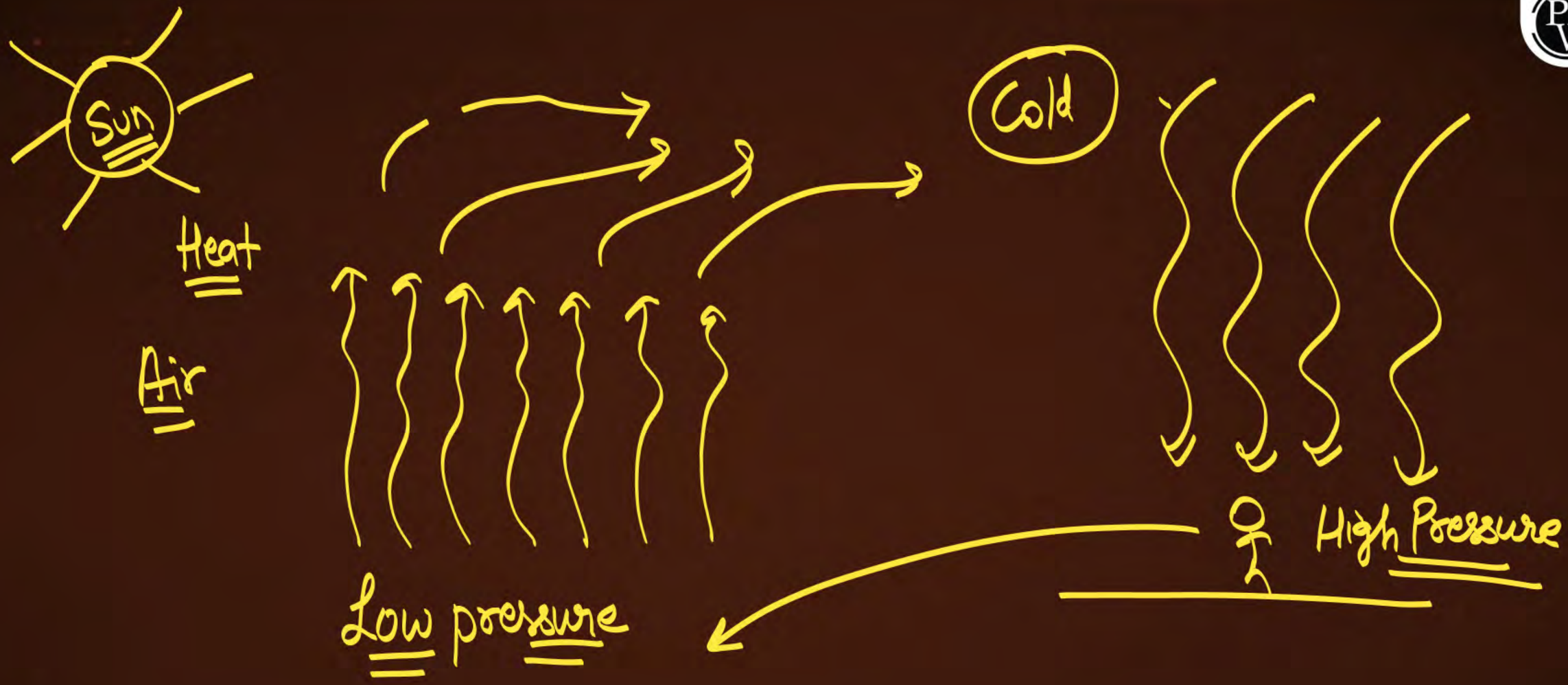
↓

Air Becomes Heavy



→ High pressure

Low ← High


Sun
Heat
Air
Cold
Low pressure
High Pressure

Distribution of Temperature



3. The temperature touches -40° C in Kargil in the month of December whereas Kerala records 20◦ C or 22° C in the same month.

4. The annual range of temperature is 3◦ C along the Malabar Coast and more than 20° C in the interiors.

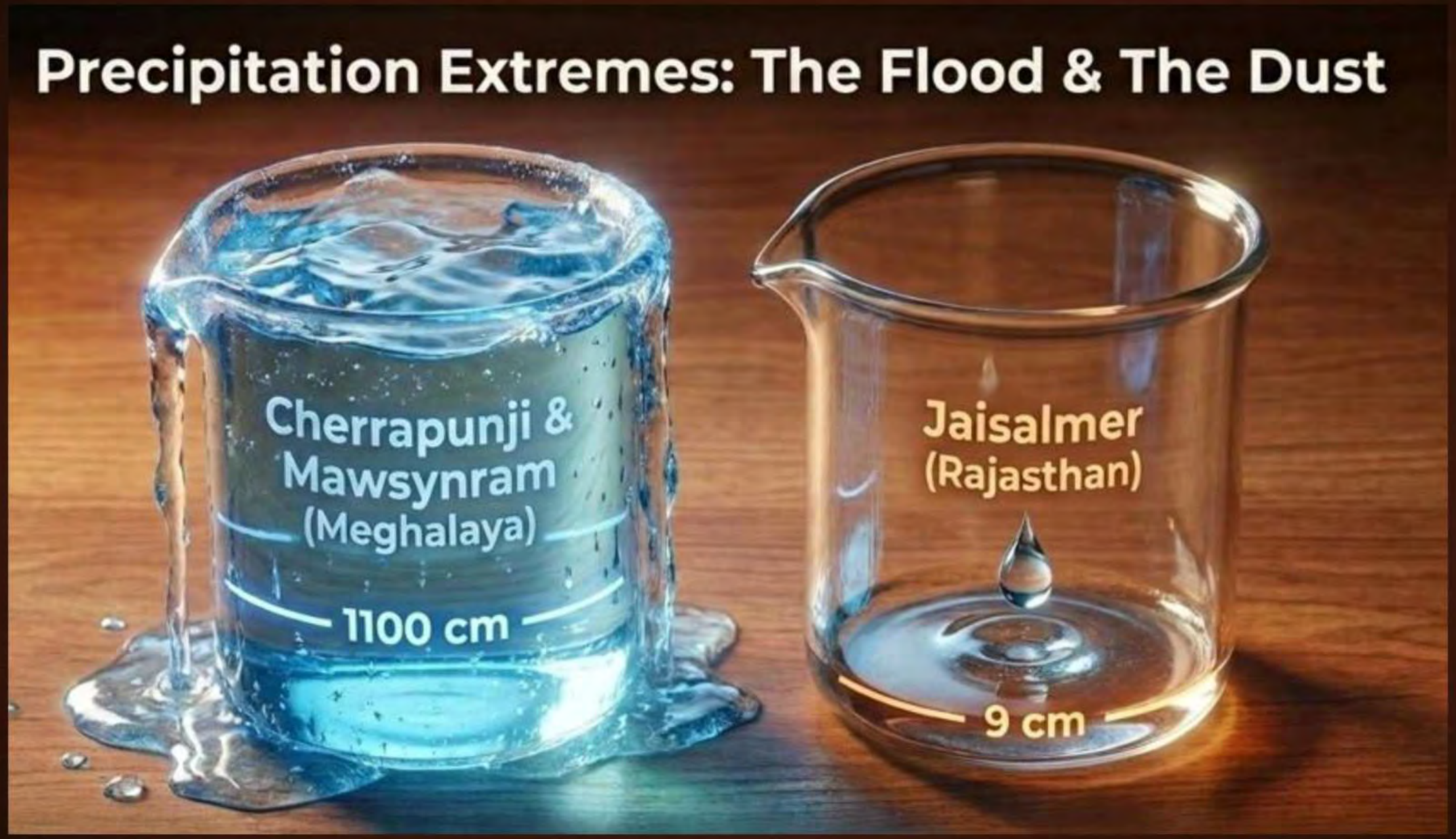
5. The difference between day and night temperatures in the Andaman Islands and Kerala is hardly seven or eight degree celsius, whereas in the Thar desert it is between 25° C to 30° C.

Precipitation



- Variations occur not only in the type of precipitation but also in its amount and the seasonal distribution.
- Snowfall occurs in the Himalayas, whereas it only rains over the rest of the country.

Precipitation Extremes: The Flood & The Dust



Precipitation



1. Cherrapunji and Mawsynram in Meghalaya get about 1100 cm rain in a year, whereas Jaisalmer in Rajasthan hardly receives 9 cm of rainfall in a year.
2. Tura in Meghalaya gets an amount of rainfall in a single day which is equal to 10 years of rainfall at Jaisalmer in Rajasthan.
3. The Coromandel Coast remains dry in the months of July and August, whereas the Ganga delta and the coastal plains of Odisha are hit by strong storms almost every third or fifth day during these months.
4. Most parts of India receive rainfall during June-September, but the coastal areas of Tamil Nadu get rains in the beginning of the winter season.

1. Which of the following latitudes passes through the middle of India, dividing it into tropical and subtropical zones?

A) Equator

B) Tropic of Capricorn

C) Tropic of Cancer

D) Arctic Circle

2. The Himalayas act as a climatic barrier by protecting India from which winds?

A) South-West Monsoon

B) Cold Siberian winds

C) Loo

D) Trade winds

3. Why does Kochi experience a lower annual range of temperature compared to Delhi?

A. Because it is further from the Equator

B. Due to its higher altitude

C. Due to its maritime influence

D. Due to heavy rainfall throughout the year

Jaiselmer → Day - 48°C, 16°C } 32°C

Kerala - Day - 32°C, Night - 27°C } 5°C

Why ??

5. Which mountain range causes a rain-shadow effect on the Deccan Plateau?

A) Aravallis

B) Himalayas

C) Eastern Ghats

D) Western Ghats

It is month of December.

Jaiselmer → Night → 4°c. → Continental

Chennai - Night → 28°c → Maritime → Land Breeze / Sea

Explain why?

Jaiselmer has Continental influence — Land < Heat Cold quickly

Kerala it has oceanic influence — Land Sea Breeze

Himalayas
(The Bouncer)

Latitude
(The Divider)



Altitude
(The Cooler)

Relief
(The Squeezer)

Summer
SW Monsoon
(The Inflow)

Winter
NE Retreat
(The Outflow)

India doesn't just have one climate; it has an intricate, interconnected ecosystem. Extreme altitudes, mountain shields, and shifting oceanic winds combine every single day to create the most dynamic weather puzzle on Earth.

NotebookLM

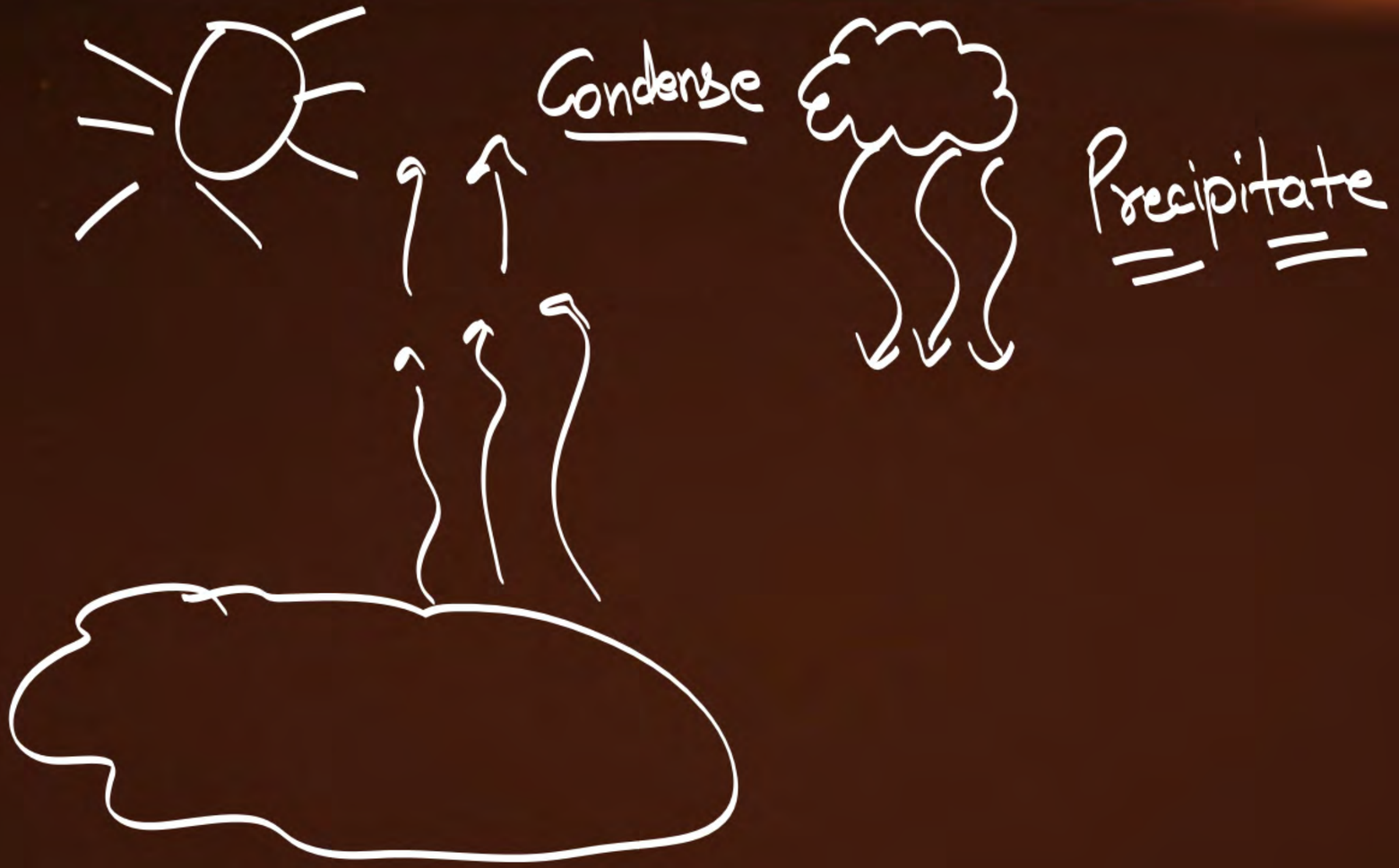
Factors Affecting the Climate of India

- Climate refers to a generalised and composite picture of the average weather conditions spread over a long period, for a given large area.

The factors which influence the climate of India are the following:

1. **The Himalayas:** The Himalayas form a climatic barrier separating the Indian subcontinent from the rest of Asia. They not only prevent the cold Siberian winds from entering the Indian region and from India becoming a cold desert but they also force the moisture laden South West Monsoon winds to shed rainfall in India or else India would be a dry region.


Orographic Rain
Step 1 : Moisture carried from the Arabian Sea.
Step 2: Windward Side: Heavy Precipitation.
Rain Shadow Area
Step 3: Leeward Side (Deccan Plateau): Moisture depleted, creating a dry rain-shadow area.


Condense
Precipitate


Precipitate
Cold
Orographic Rain
Rainfall Caused due to presence of mountains
No moisture
Rain Shadow Area

Factors Affecting the Climate of India

→ Physical features of any place

Relief and rainfall: The windward side of Western Ghats receive rainfall from the S-W Monsoon and the windward side of Eastern Ghats receive rainfall from N-E Monsoon.





Peninsula

↳ An area Surrounded By water on 3 Sides.

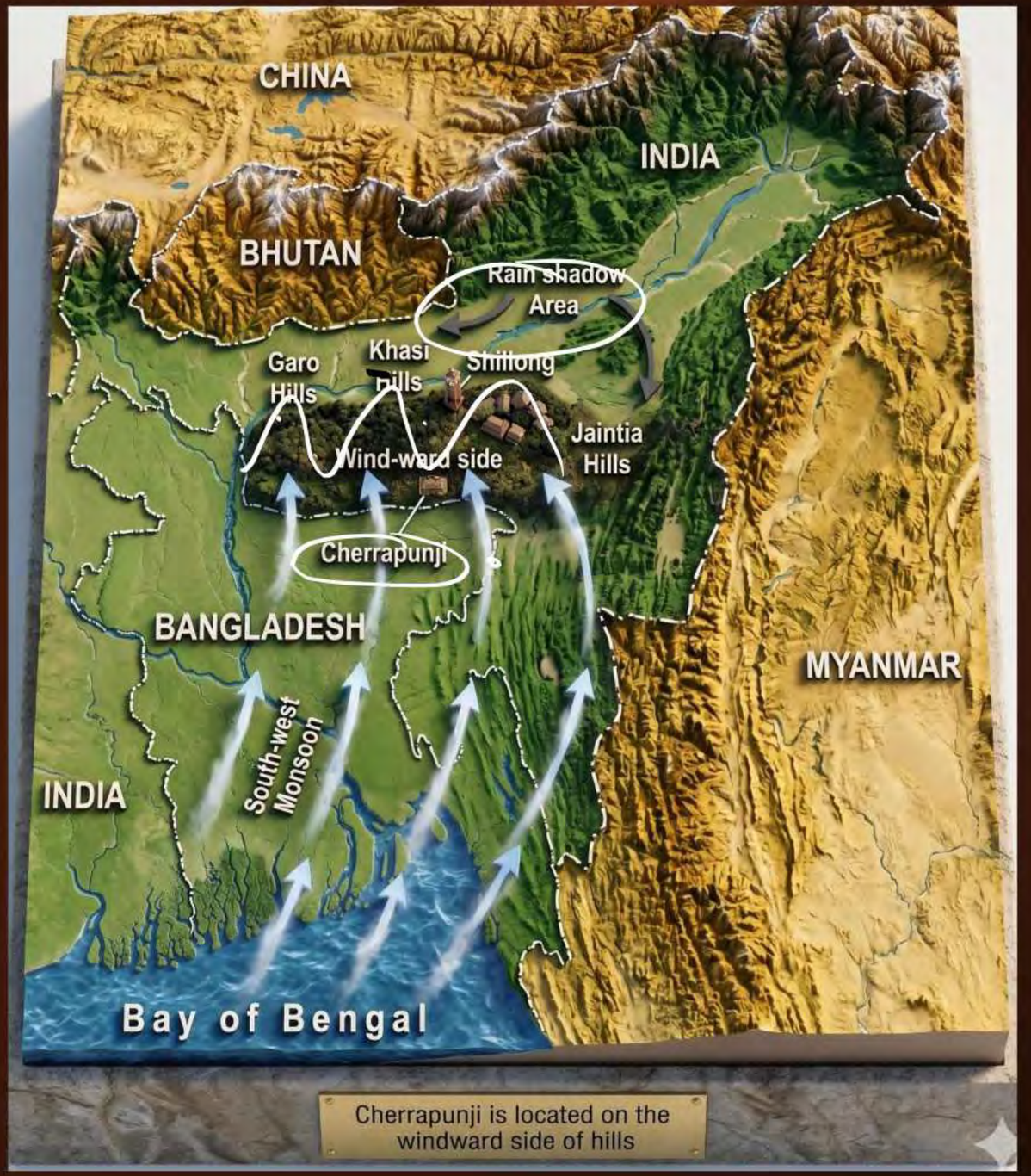
Factors Affecting the Climate of India

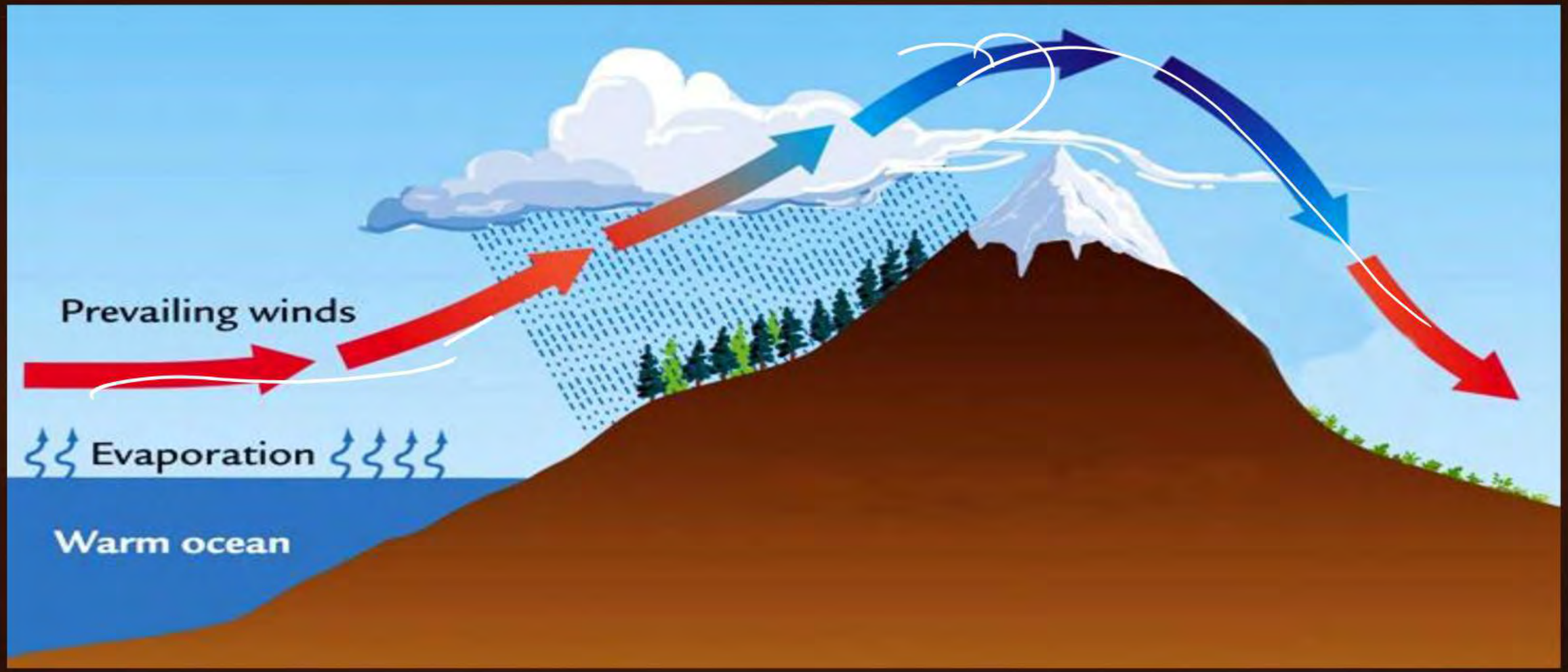
2. **Varied Relief:** Relief plays an important role in the climatic conditions of India. The Western Ghats get heavy rainfall on the western side because they stand in the way of South West Monsoon winds which come from the Arabian Sea.

- On the other hand, the Deccan Plateau gets less rainfall as it lies in the rain-shadow area of the Western Ghats.






CHINA
INDIA
BHUTAN
Rain shadow Area
Garo Hills
Khasi Hills
Shillong
Jaintia Hills
Wind-ward side
Cherrapunji
BANGLADESH
MYANMAR
South-west Monsoon
INDIA
Bay of Bengal
Cherrapunji is located on the windward side of hills


Prevailing winds
Evaporation
Warm ocean

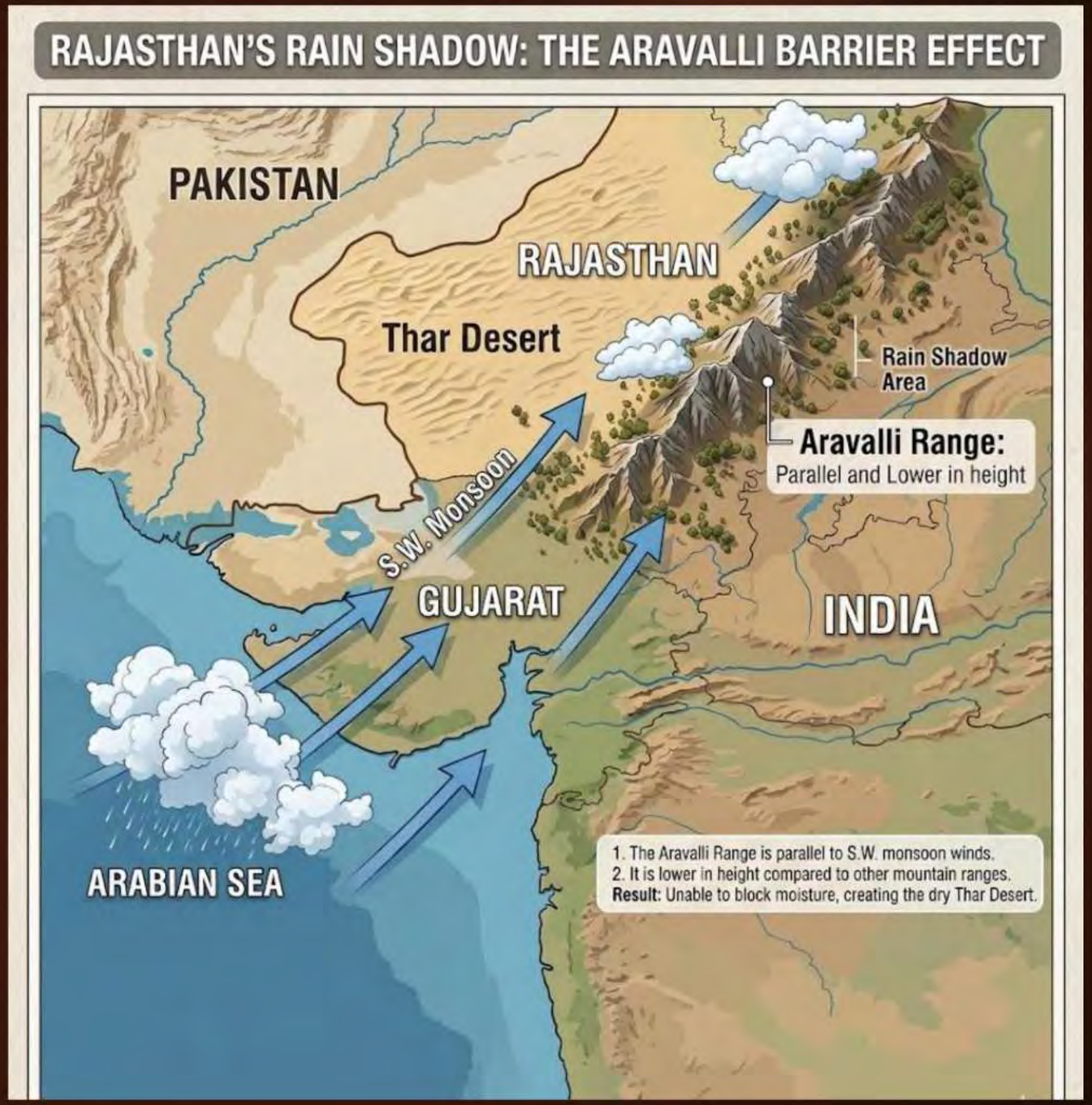
Factors Affecting the Climate of India

- In Rajasthan, the Aravalli Range is parallel to the direction of the South West Monsoon winds. So it is unable to stop the moisture laden winds. This makes Rajasthan a dry area.

- The southern hills of Assam get heavy rainfall but the northern ones do not. This is because the southern hills force the moisture laden winds to shed their moisture before they proceed northwards.

No Rainfall in Rajasthan

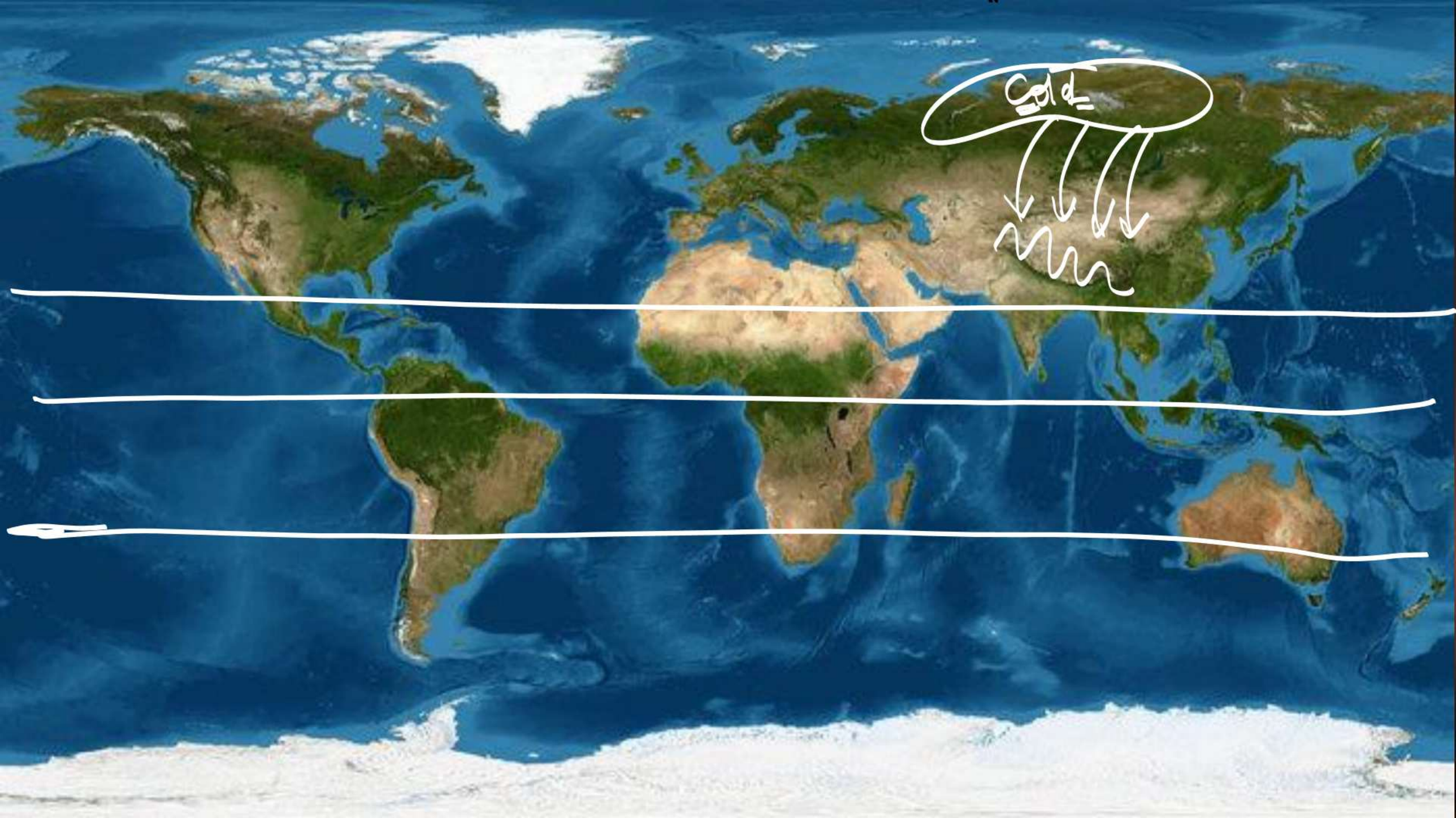






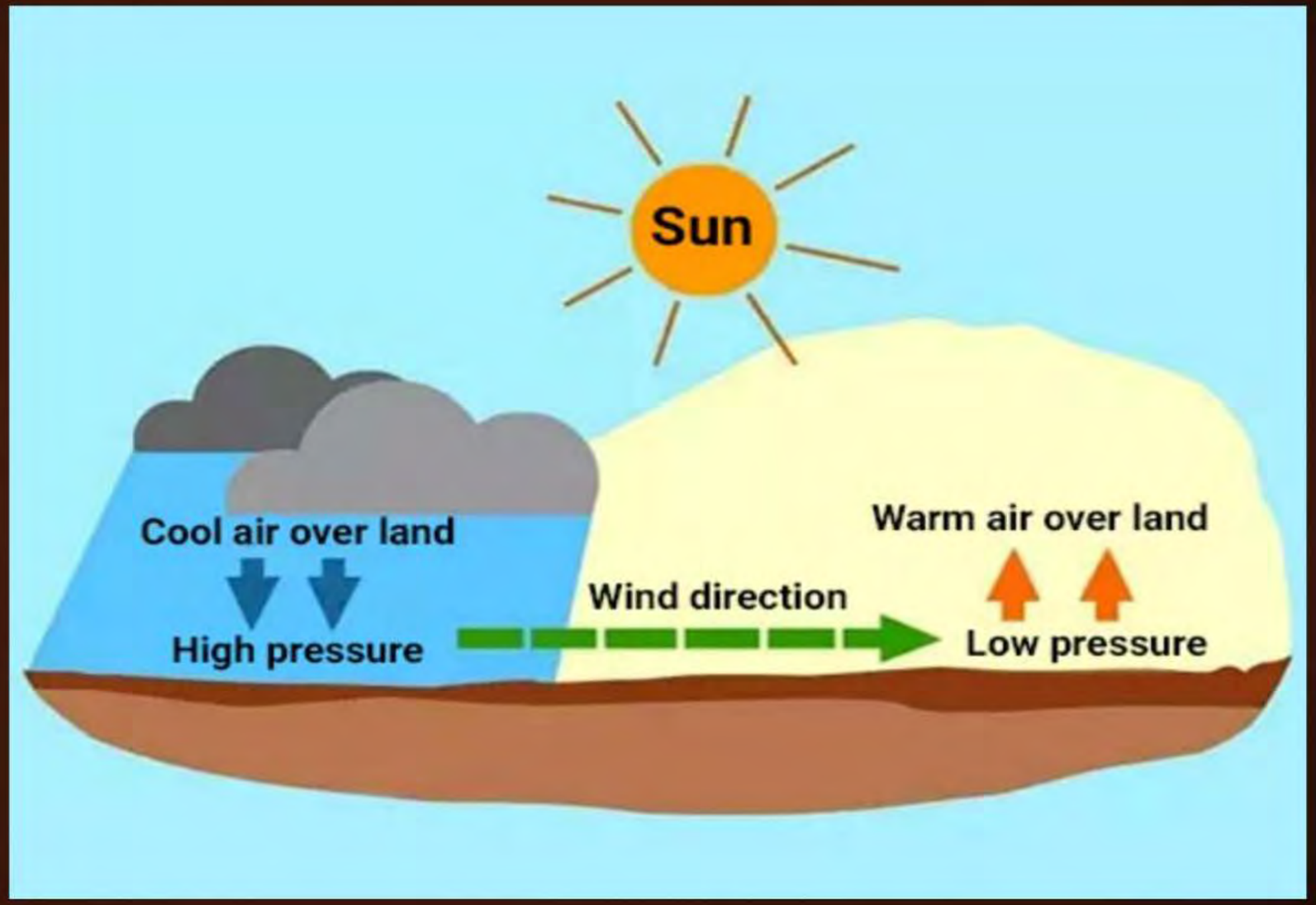
Factors Affecting the Climate of India

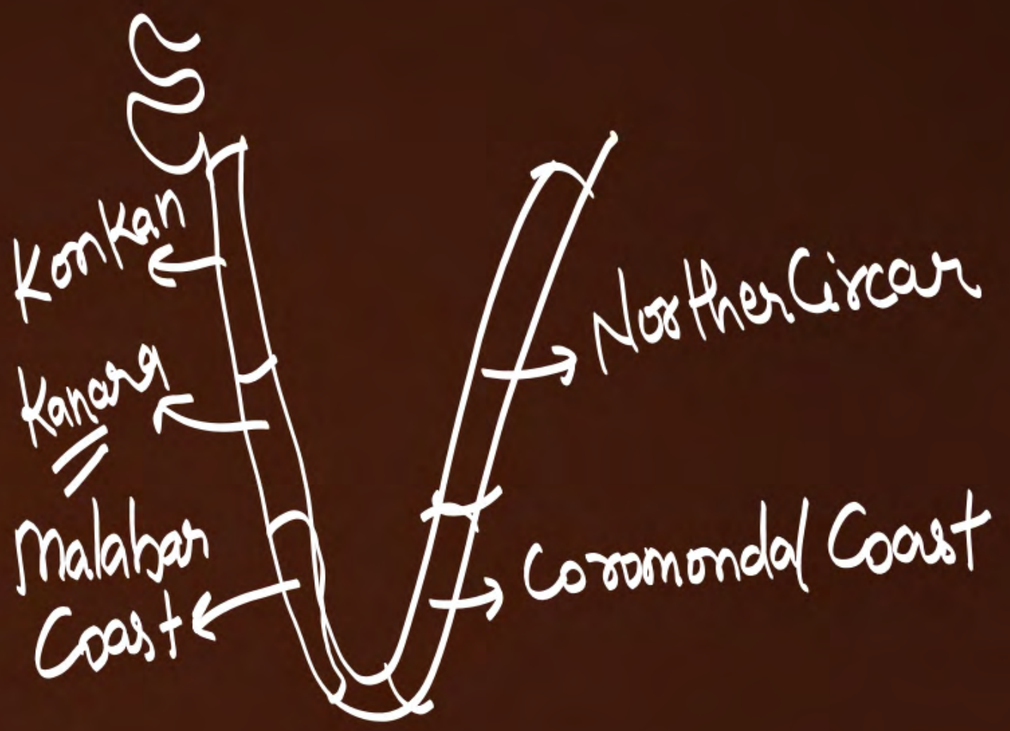
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Cold

Movement of Air

High — Low




Konkan
Kanara
Malabar Coast
Northern Circar
Coromondal Coast

Thank You